

Chapter 1 — Characteristics of good datasets

Judge quality as a bundle of task-relative dimensions

Learning objectives

- List five quality dimensions from the chapter
- Diagnose accuracy, completeness, and consistency defects
- Explain why metadata and documentation matter

Five quality dimensions

- **Accuracy** — values reflect ground truth closely enough
- **Completeness** — needed records and attributes are present
- **Consistency** — uniform formats, units, definitions
- **Relevance** — data answers the intended question
- **Timeliness** — data is sufficiently up to date

Example 1.11 — Incorrect transaction

```
$ cd modules/chapter1/example11/ && bash run.sh
```

Completeness and consistency

- Missing age or diagnosis fields bias clinical models
(eg:1.12)
- Mixed date formats break chronological sorting
(eg:1.13)
- Defects interact: complete yet inaccurate still fails the task
- Systematic repair is deferred to later chapters

Metadata and documentation

```
$ cd modules/chapter1/example14/ && bash run.sh
```

Takeaways

- Quality is multi-dimensional and task-relative
- Spot accuracy, missingness, and format clashes early
- Metadata + docs make datasets usable beyond their author